

Classification: Logistic Regression, ROC, and Naive Bayes

Biostatistics 735/794

September 17, 2026

MLE Logistic Regression

- ▶ In logistic regression, β can be estimated by maximizing the log-likelihood

$$\log\{L(\beta)\} = \sum_{i=1}^N \log \{ \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \}$$

- ▶ This can be written in terms of $\beta = \{\beta_{01}, \beta_1\}$ as

$$\log\{L(\beta)\} = \sum_{i=1}^N \left\{ y_i \beta' \mathbf{x}_i - \log(1 + e^{\beta' \mathbf{x}_i}) \right\}$$

where $\mathbf{x}$ now contains an intercept term.

- ▶ β is usually estimated using an algorithm called *iteratively reweighted least squares* or IRLS.

L_1 penalized logistic regression

- ▶ The L_1 Lasso penalty discussed previously can be used for variable selection and shrinkage.
- ▶ For the 2-class case, we look to find

$$\max_{\beta} \left[\sum_{i=1}^N \left\{ y_i \beta' \mathbf{x}_i - \log(1 + e^{\beta' \mathbf{x}_i}) \right\} - \lambda \sum_{j=1}^p |\beta_j| \right]$$

- ▶ This is a difficult concave optimization, but the IRLS solution can be applied.
- ▶ This leads to a repeated application of a weighted lasso algorithm.

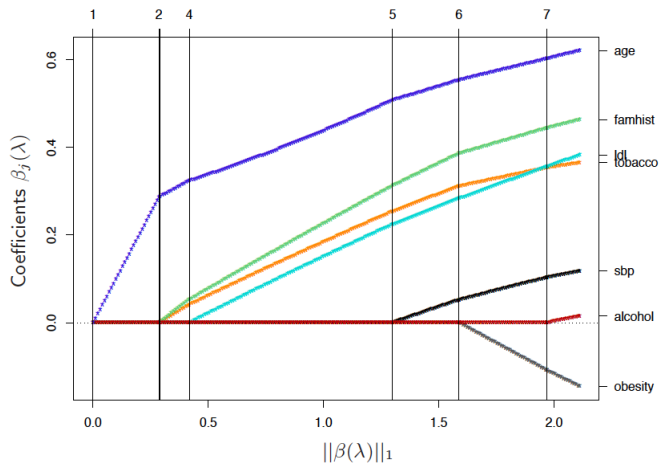


Figure: From ESL (online version). Note that the paths are not linear.

Linear Discriminant Analysis (LDA): Setup

- ▶ Suppose we have data from K groups (e.g., smokers vs. nonsmokers).
- ▶ We model predictors X as coming from a multivariate normal (MVN) distribution for each group:

$$X \sim MVN(\mu_k, \Sigma_k), \quad k = 1, \dots, K.$$

- ▶ If all groups share the same covariance matrix ($\Sigma_k = \Sigma$), then the boundaries between groups turn out to be **linear**.
- ▶ This is the key idea of **Linear Discriminant Analysis (LDA)**.

Linear Discriminant Function

- ▶ LDA builds a **discriminant function** for each group:

$$\delta_k(x) = x' \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k' \Sigma^{-1} \mu_k + \log \pi_k$$

where

- ▶ μ_k = mean vector for group k ,
 - ▶ π_k = prior probability of being in group k ,
 - ▶ Σ = common covariance matrix.
- ▶ The classification rule:

Assign x to the group with the largest $\delta_k(x)$.

- ▶ In practice: this means each group is separated by a **linear boundary**.

Linear Discriminant Analysis Example

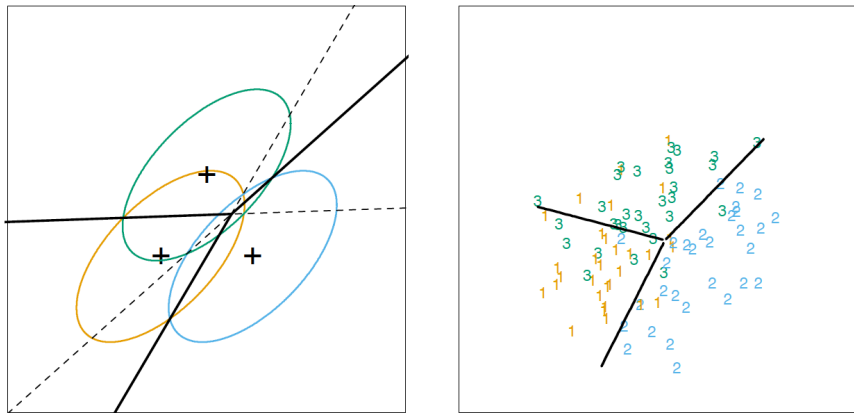


Figure: From ESL (online version).

Estimating the Parameters in LDA

- ▶ Estimate group priors: $\hat{\pi}_k = N_k/N$.
- ▶ Estimate group means: $\hat{\mu}_k = \text{average of } x_i \text{ in group } k$.
- ▶ Estimate pooled covariance: $\hat{\Sigma} = \text{weighted average of within-group covariance}$.
- ▶ Note:
 - ▶ LDA does not rely heavily on the normality assumption for prediction.
 - ▶ The “cutpoint” (intercept) is more sensitive to distributional assumptions — often chosen or checked with cross-validation.

Quadratic Discriminant Analysis (QDA)

- ▶ If we allow each group to have its own covariance matrix (Σ_k), then the discriminant function becomes **quadratic**.
- ▶ The classification rule is still to assign x to the group with the largest $\delta_k(x)$, but the boundary curves can now be **non-linear**.
- ▶ Compared to LDA:
 - ▶ More flexible, captures complex separation between groups.
 - ▶ Requires estimating more parameters, so can overfit with small data.

LDA vs QDA: A Tradeoff

- ▶ **LDA**: simpler, linear boundaries, fewer parameters. Works well when groups are roughly separated by lines or planes.
- ▶ **QDA**: more flexible, quadratic boundaries, more parameters. Works well when groups need curved boundaries to be separated.
- ▶ Which to choose?
 - ▶ Try both and compare using cross-validation.
 - ▶ LDA is often more stable when sample size is small.

Naive Bayes: Key Idea

- ▶ Uses Bayes' Theorem:

$$\Pr(G = k | X = x) \propto \Pr(G = k) f_k(x)$$

- ▶ Challenge: estimating the joint density $f_k(x)$ for high-dimensional data.
- ▶ **Naive Bayes assumption:** Features are conditionally independent given the class:

$$f_k(x) = \prod_{j=1}^p f_{kj}(x_j)$$

- ▶ Result: Simple, fast, and effective even in high-dimensional data (e.g., text classification, genetics, EHR data).

From Posterior to Classification

- ▶ After estimating class priors $\pi_k = \Pr(G = k)$ and feature distributions $f_{kj}(x_j)$:

$$\Pr(G = k|X = x) = \frac{\pi_k \prod_{j=1}^p f_{kj}(x_j)}{\sum_{\ell} \pi_{\ell} \prod_{j=1}^p f_{\ell j}(x_j)}$$

- ▶ For a new subject with features x , we compute the posterior probability for each class k .
- ▶ **Classification rule:** Assign x to the class with the largest posterior probability.

$$\hat{G}(x) = \arg \max_k \Pr(G = k|X = x)$$

Estimation Details

The first component to estimate is the **class priors** (π_k). This can be estimated in two ways:

1. MLE

$$\hat{\pi}_k = \frac{N_k}{N}$$

where N_k is the number of samples in class k .

- ▶ Simple and unbiased (for large N).

2. Smoothing (Laplace / Dirichlet priors):

$$\hat{\pi}_k = \frac{N_k + \alpha}{N + \alpha}$$

- ▶ Prevents zero probabilities for rare/unseen features.
- ▶ Slight bias, but lower variance and improved robustness.

Estimation details (cont.)

- ▶ The second component to estimate is the **class-conditional feature distributions** ($f_{kj}(x_j)$):

$$f_k(x) = \prod_{j=1}^p f_{kj}(x_j)$$

$f_{kj}(x_j)$ - the probability density or mass function of x_j for class k (i.e., the likelihood of the mean radius for malignant tumors)

- ▶ Different feature types are modeled with different conditional distributions:
 - ▶ Binary feature (e.g., smoker yes/no) $\rightarrow f_{kj}(x_j)$ is Bernoulli.
 - ▶ Count feature (e.g., number of visits) $\rightarrow f_{kj}(x_j)$ is Multinomial (or Poisson).
 - ▶ Continuous feature (e.g., BMI) $\rightarrow f_{kj}(x_j)$ is Gaussian.

Estimating Class-Conditional Distributions

► Bernoulli Naive Bayes (binary features):

- Each feature $X_j \in \{0, 1\}$.
- For class k , estimate success probability:

$$\hat{p}_{kj} = \frac{\#\{x_j = 1 \text{ in class } k\}}{N_k}$$

- Likelihood term: $f_{kj}(x_j) = \hat{p}_{kj}^{x_j}(1 - \hat{p}_{kj})^{1-x_j}$.
 - Problem: assigns zero probability if a feature never appears in a class. Smoothing (discussed above) is commonly used.
- Note: Bernoulli Naive Bayes is a procedure when all X are Bernoulli.

Estimating Class-Conditional Distributions

► Multinomial Naive Bayes (counts):

- Each feature $X_j = (x_{j1}, x_{j2}, \dots, x_{jL})$ is a count (e.g., word counts) or category where $x_{j\ell} = 1$ if in category ℓ .
- Estimate probabilities by normalized counts:

$$\hat{p}_{kj\ell} = \frac{n_{kj\ell}}{\sum_m n_{kjm}}$$

where $n_{kj\ell}$ is the total count of feature j category ℓ in class k .

- Likelihood term: $f_{kj}(x_j) \propto \prod_{\ell=1}^L \hat{p}_{kj\ell}^{x_{j\ell}}$.
 - Smoothing (discussed above) is commonly used.
- Note: Multinomial Naive Bayes is a procedure when all X are multinomial.

Estimating Class-Conditional Distributions

- ▶ **Continuous features (Gaussian):**

- ▶ Assume $X_j \sim \mathcal{N}(\mu_{kj}, \sigma_{kj}^2)$ in class k .
- ▶ Estimate mean and variance from class-specific data.

- ▶ **Continuous features (Nonparametric):**

- ▶ Use nonparametric density estimation methods that don't assume any particular form.

Combining Class-Conditional Distributions

- ▶ Mixed data types can be combined in one model.

$$f_k(\mathbf{x}) = \prod_{j=1}^p f_{kj}(x_j)$$

- ▶ For example, suppose $\mathbf{x} = (x_1, x_2, x_3)$ which are binary, categorical, and continuous variables, respectively.
- ▶ Then $f_k(\mathbf{x})$ can multiply a Bernoulli (f_{k1}), Multinomial (f_{k2}), and the Gaussian (f_{k3}) likelihoods together.

$$f_k(x_1, x_2, x_3) = f_{k1}(x_1)f_{k2}(x_2)f_{k3}(x_3)$$

- ▶ Bernoulli, Multinomial, and Gaussian are not separate methods—they are **choices for feature distributions** within the same Naive Bayes framework.

Naive Bayes: Advantages and Disadvantages

Advantages

- ▶ **Simple and fast** to fit and predict
- ▶ Works well with **high-dimensional** data
- ▶ Requires relatively **little training data**
- ▶ Can work well when predictors are approximately independent
- ▶ Produces **probabilistic predictions**

Disadvantages

- ▶ Assumes predictors are **conditionally independent given the class***
- ▶ This assumption is often unrealistic
- ▶ Correlated predictors can lead to **poor probability estimates**
- ▶ Continuous predictors require additional distributional assumptions
- ▶ Less flexible than methods that model relationships among predictors

*Tree-Augmented Naive Bayes (TAN) relaxes the independence assumption.

Adding a Cost Function

- ▶ Standard classifiers minimize 0–1 loss:

$$\text{Loss}(\hat{y}, y) = I(\hat{y} \neq y)$$

- ▶ In practice, false positives and false negatives may not be equally costly.
- ▶ Incorporate a cost matrix $C(k, \ell)$:

$$\hat{y} = \arg \min_k \sum_{\ell} C(k, \ell) \Pr(G = \ell | X = x)$$

- ▶ Examples:
 - ▶ Cancer screening: missing a true case (false negative) may have much higher cost than a false alarm.
 - ▶ Predicting hospital readmission: over-predicting (false positives) may waste resources.
- ▶ Cost-sensitive classification aligns model decisions with public health priorities.

Summary

- ▶ Naive Bayes simplifies high-dimensional classification by assuming conditional independence.
- ▶ Variants (Bernoulli, Multinomial) adapt to different data types.
- ▶ Cost-sensitive classifiers allow tailoring decisions to real-world consequences.

Logistic Regression with GAMs

- ▶ Standard logistic regression assumes:

$$\log \frac{\pi(x)}{1 - \pi(x)} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

- ▶ Predictors enter the model linearly, which may be too restrictive.
- ▶ Example: effect of age on disease risk may be non-linear (e.g., flat at younger ages, steep increase after midlife).
- ▶ **Generalized Additive Models (GAMs)** extend logistic regression by allowing smooth, non-linear functions of predictors:

$$\log \frac{\pi(x)}{1 - \pi(x)} = \beta_0 + f_1(x_1) + f_2(x_2) + \cdots + f_p(x_p)$$

How GAMs Work

- ▶ Each $f_j(\cdot)$ is a **smooth function** (often splines) estimated from the data.
- ▶ The model still adds these terms together (hence “additive”), keeping interpretation manageable.
- ▶ Smoothness is controlled by a penalty:
 - ▶ Too smooth $\Rightarrow$ miss important patterns.
 - ▶ Too wiggly $\Rightarrow$ risk of overfitting.
- ▶ Example formula in R:

```
gam(y ~ s(age) + bmi + s(bp), family = binomial)
```

- ▶ Output: predicted probability of outcome with flexible, data-driven shapes for age and blood pressure.

Why Use GAMs in Public Health?

- ▶ Captures complex, realistic exposure–response relationships.
- ▶ Provides smoother and more interpretable fits compared to categorizing continuous variables into bins.
- ▶ Example applications:
 - ▶ Smoking intensity and lung cancer risk.
 - ▶ Air pollution exposure and probability of asthma diagnosis.
 - ▶ BMI and probability of diabetes onset.
- ▶ Visualizations of $f_j(x)$ give insight into how predictors affect the probability of disease.